Pixel-Space One-Step Denoiser Matching for Cross-VAE On-Policy Distillation

Working Analysis (Draft)

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Abstract

This working note formalizes a concrete cross-VAE distillation scheme: the student rolls out a clean latent, decodes it to an image; that image is re-encoded by *each* model's own VAE, re-noised to a common level t, denoised in *one step* by that model, and decoded back to pixels; the pixel-space discrepancy between the teacher's and the student's reconstructions is the training signal. We show this is a symmetric, fully pixel-anchored instance of the "shared-space" route (companion note `avoid_inverse_transport_analysis.tex`, Route B): it needs *no* latent transport map T or its inverse. We characterize what it distills (one-step x_0-prediction matching across noise levels $\Leftrightarrow$ velocity/score matching $\Leftrightarrow$ probability-flow matching, transported through the decoders), its gradient structure (only adjoints/VJPs of forward maps), its behavior across t, and its irreducible reconstruction floor. We also contrast it with DMD.

1 Setup

Two frozen VAEs give encoders/decoders $(\mathcal{E}_A, \mathcal{D}_A)$ on teacher latent space $\mathcal{Z}_A \subseteq \mathbb{R}^{d_A}$ and $(\mathcal{E}_B, \mathcal{D}_B)$ on student latent space $\mathcal{Z}_B \subseteq \mathbb{R}^{d_B}$, over a shared pixel space $\mathcal{X}$. The teacher velocity v_A (frozen) and the student velocity v_θ each induce a *one-step clean-sample estimator* (denoiser). With a noising schedule $z^t = \alpha_t z^0 + \sigma_t \varepsilon$ ($\alpha_0 = 1, \sigma_0 = 0$, $\varepsilon \sim \mathcal{N}(0, I)$), the denoiser of a model with velocity v is the affine image of its velocity,

$$\mathrm{den}(z^t, t) \;:=\; \widehat{z^0}(z^t, t) \;=\; (\text{schedule-dependent affine map of } v(z^t, t)), \tag{1}$$

i.e. the model's "one step to clean" prediction. Write den_A for the teacher and den_θ for the student. We assume $d_A \geq d_B$, but the scheme below does not use this.

2 The Scheme

Definition 1 (Pixel-space one-step denoiser matching)**.** Let the student roll out a clean latent $z_B^0 \sim \rho_\theta$ and decode it to an image

$$x \;:=\; \mathcal{D}_B\big(z_B^0\big) \;\in\; \mathcal{X} \qquad \text{(on-policy image)}. \tag{2}$$

For a noise level t and independent draws $\varepsilon_A \in \mathbb{R}^{d_A}$, $\varepsilon_B \in \mathbb{R}^{d_B}$, define the two *decode-of-one-step-denoise* branches

$$\text{teacher:} \quad y_A(x, t, \varepsilon_A) \;:=\; \mathcal{D}_A\Big(\mathrm{den}_A\big(\alpha_t\, \mathcal{E}_A(x) + \sigma_t \varepsilon_A,\ t\big)\Big), \tag{3}$$

$$\text{student:} \quad y_B(x, t, \varepsilon_B) \;:=\; \mathcal{D}_B\Big(\mathrm{den}_\theta\big(\alpha_t\, \mathcal{E}_B(x) + \sigma_t \varepsilon_B,\ t\big)\Big). \tag{4}$$

The training objective is the pixel-space discrepancy

$$L(\theta) \;=\; \mathbb{E}_{x \sim \rho_\theta}\, \mathbb{E}_{t \sim p(t)}\, \mathbb{E}_{\varepsilon_A, \varepsilon_B}\Big[\, w(t)\, \big\|\, \mathrm{sg}\big[y_A(x, t, \varepsilon_A)\big] - y_B(x, t, \varepsilon_B)\, \big\|^2\, \Big], \tag{5}$$

with a time weight $w(t)$ and a stop-gradient $\mathrm{sg}[\cdot]$ on the teacher target.

Observation 1 (No transport map is needed; pixels are the shared anchor)**.** Both branches only ever apply *forward* maps within a single latent space ($\mathcal{E}_\bullet$, $\mathrm{den}_\bullet$, $\mathcal{D}_\bullet$), and meet in the shared pixel space $\mathcal{X}$. The two latent spaces are never put into correspondence: there is no $\mathsf{T} : \mathcal{Z}_A \to \mathcal{Z}_B$ and, in particular, no underdetermined inverse $\mathsf{T}^{-1} : \mathcal{Z}_B \to \mathcal{Z}_A$. The cross-VAE mismatch is dissolved by comparing *behavior in pixel space* rather than *coordinates in latent space.* This is the symmetric, both-sides-decoded form of Route B of the companion note.

3 What the Objective Distills

Proposition 1 (Denoiser family $\Leftrightarrow$ velocity $\Leftrightarrow$ flow)**.** For a fixed schedule, the one-step denoiser family $\{\mathrm{den}(\cdot,t)\}_{t\in(0,1)}$ is in bijection with the velocity family $\{v(\cdot,t)\}_{t\in(0,1)}$ via the affine relation (1); the velocity family determines the probability-flow ODE and hence the model's clean marginal p^0. Consequently, matching one-step x_0-predictions *at all* t (within a common space) is equivalent to matching velocities/scores, i.e. to matching the generated distribution.

Remark 1 (The pixel version = velocity matching transported through the decoders)**.** Objective (5) compares $\mathcal{D}_A(\mathrm{den}_A(\cdot))$ against $\mathcal{D}_B(\mathrm{den}_\theta(\cdot))$. Were the decoders invertible and identical, this would reduce (by Prop. 1) to exact latent denoiser/velocity matching. In general it is velocity matching *conjugated by the decoders*: the comparison is transported to pixels by the *forward* maps $\mathcal{D}_A, \mathcal{D}_B$ instead of by a latent transport map. This is precisely why the ill-posed vector-field pushforward (and its section/inverse) is avoided—see companion note, Prop. "projectability".

Proposition 2 (Fixed point)**.** Suppose $\mathcal{D}_B$ is injective on the relevant support and $w(t) > 0$. If $L(\theta)$ attains its θ-dependent minimum with $y_B \equiv y_A$ (in expectation over ε) for all $x \in \mathrm{supp}(\rho_\theta)$ and all t, then den_θ reproduces the pixel-conjugated teacher denoiser on that support, hence (Prop. 1) the student's probability flow matches the teacher's *as observed through* $\mathcal{D}_B$. As ρ_θ then covers the teacher's image manifold, the match is self-reinforcing—the standard on-policy fixed point.

4 Gradient Structure and Stop-Gradient Design

The student parameter θ enters (5) through (i) the on-policy image x (produced by the rollout v_θ) and (ii) the student branch y_B (the denoiser being trained). The teacher branch y_A depends on θ only through x.

Remark 2 (Recommended gradient paths)**.** Treat x as a *sampled on-policy state* (detach it) and stop-gradient the teacher target; then the gradient flows only through the student branch,

$$\nabla_\theta L = \mathbb{E}\Big[-2\,w(t)\,\big(\mathrm{sg}[y_A] - y_B\big)^\top \mathbf{J}_{\mathcal{D}_B}\,\mathbf{J}_{\mathrm{den}_\theta}\,\partial_\theta \mathrm{den}_\theta\Big], \tag{6}$$

which uses only *adjoints* (vector–Jacobian products) of the forward maps $\mathcal{D}_B$ and den_θ; no map is inverted. Crucially, the rollout policy and the denoiser are the *same* network v_θ, so regressing the denoiser on on-policy states (6) *also* improves the rollout—this is the on-policy distillation mechanism. Propagating additionally through x (the rollout) is possible but expensive and can create a trivial fixed point (the student may prefer images on which the two VAEs happen to agree); detaching x avoids this.

5 Behavior Across Noise Levels and the Reconstruction Floor

Remark 3 (Limiting behavior in t)**.** At $t \to 0$: $\mathrm{den}_\bullet \to \mathrm{id}$, so $y_A \to \mathcal{D}_A\mathcal{E}_A(x)$ and $y_B \to \mathcal{D}_B\mathcal{E}_B(x)$; the loss tends to the VAE round-trip discrepancy, largely θ-independent and *uninformative.* At $t \to 1$: the branches compare the two models' *generative priors* (one-step guesses from near-noise); the distillation signal is strongest but has the highest variance. A weight $w(t)$ emphasizing mid-to-high t is therefore natural.

Proposition 3 (Irreducible floor from VAE mismatch)**.** Even a student whose implied clean distribution equals the teacher's incurs a nonzero loss, because y_A and y_B pass through *different* decoders. Define the reconstruction floor

$$\delta_{\mathrm{rec}} \;:=\; \mathbb{E}_{x\sim\rho_\theta}\; \|\mathcal{D}_A\mathcal{E}_A(x) - \mathcal{D}_B\mathcal{E}_B(x)\|\,, \tag{7}$$

the pixel gap between the two VAEs' reconstructions of the same image. Then (5) is bounded below by a δ_{rec}-type term (exactly δ_{rec}^2 in the $t \to 0$ limit) and should be interpreted *modulo* this floor; e.g. one may subtract the student's own reconstruction baseline to debias.

6 Relationship to DMD, and to the Companion Routes

Remark 4 (DMD comparison)**.** Like DMD/VSD, the states probed are *noised outputs of the student generator* (here: noised re-encodings of $x \sim \rho_\theta$), so the scheme is on-policy w.r.t. the student's *output* distribution (not its trajectory states). Unlike DMD, the signal is a *direct pixel-space regression* between decoded one-step denoisings (5), back-propagated through the student denoiser; it needs *no* auxiliary online "fake-score" network and does not form the (real score − fake score) gradient. It is thus closer to a denoiser/x_0-matching distillation than to variational score distillation, at the cost of back-propagating through $\mathcal{D}_B$ each step.

Remark 5 (Placement among the routes)**.** This scheme is Route B (pixel adjoint) made symmetric and swept over t: both sides do encode→noise→one-step→decode, and the target uses only the *frozen* $\mathcal{E}_A, \mathcal{D}_A$ (never a learned low→high map). It preserves the student as a velocity/denoiser network v_θ (a multi-step flow), in contrast to Route A (DMD), which naturally yields a few-step generator.

7 Assumptions and Limitations

- **Metric.** The pixel-space ℓ_2 metric, pulled back through $\mathcal{D}_B$, induces a non-Euclidean effective metric on $\mathcal{Z}_B$ (weighted by $\mathbf{J}_{\mathcal{D}_B}^\top \mathbf{J}_{\mathcal{D}_B}$); perceptual/latent structure is only implicitly respected.
- **Floor.** δ_{rec} (7) sets a noise floor; small-t terms are dominated by it and should be down-weighted or debiased (Prop. 3).
- **Variance.** $\varepsilon_A, \varepsilon_B$ are independent (dimensions differ), adding variance; averaging over several draws, or using expected (denoiser) forms, reduces it.
- **Cost.** Each step needs a teacher forward plus $\mathcal{E}_A, \mathcal{D}_A$ for the target and a differentiable pass through $\mathcal{E}_B, \mathrm{den}_\theta, \mathcal{D}_B$ for the student branch.
- **On-policy notion.** On-policy is w.r.t. the student *output* distribution (noised), not the multi-step *trajectory* states of XOPD's L1.
- **One-step coarseness.** At high t the one-step x_0-prediction is a coarse estimate for both models; the signal is meaningful only when integrated across t (Prop. 1).

8 Summary

The scheme distills the teacher's denoiser (equivalently its velocity/flow) into the student by comparing *decoded one-step denoisings in pixel space*, on the student's own on-policy image distribution, across noise levels. Its defining virtue is that pixel space serves as the shared anchor, so it requires *no* latent transport map and *no* underdetermined student$\to$teacher inverse; its gradient uses only adjoints of forward maps. Its principal caveats are the VAE-mismatch floor δ_{rec}, the pixel-metric bias, and the per-step decode cost. It keeps the student a multi-step flow model (unlike DMD) and needs no auxiliary score network (unlike VSD).